Course notes

Deriving a compound

If u is derivable on I and v Derivable over an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The fields must be checked prior to calculation.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad (\frac{1}{u})' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is derivable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines changes in f; the sign of f do not determine them.

Convexity

A function f is convex over an interval I if his curve is under his ropes: $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a two-fold detritable function, equivalences are allowed: f convex; f' increasing; $f'' \geq 0$; the curve is above all its tangents. Inequalities are reversed for a function Concave. A point of inflection is a point where the curve changes convexity.

Demonstration: position relative to tangents. Suppose $f'' \geq 0$. Fixes a and positons $g(x) = f(x) - f(a) - f'(a)(x-a)$. Like f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g reached its minimum $g(a) = 0$ in a, therefore $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not an apse of inflection. x^3 exchange rate of convexity in 0. The convexity of the exponential and its tangent in 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous in a if $\lim_{x \to a} f(x) = f(a)$. Any drifting function is continuous. Reciprocality is false: $|x|$ is continuous in 0 but not driftable in 0.

Theorem of intermediate values

If f is continuous on $[a,b]$, any real between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is also strictly monotonous, this preimage is unique.
work on an open interval using boundary at bounds.

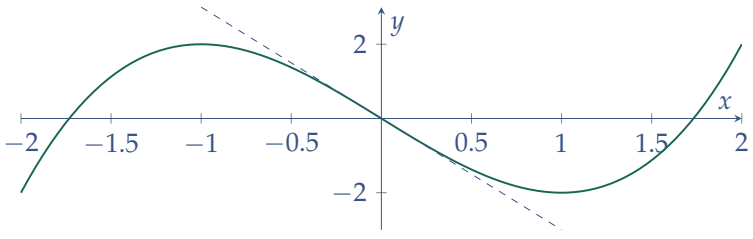
Prepared example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Like $f(0) = -1$ and $f(1) = 1$, the VIT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it's unique. $f(0{,}68) < 0 < f(0{,}69)$ provides a framework, without replacing justifications of existence and uniqueness.

A function defined by pieces. For $x \neq 1$, pose $g(x) = \frac{x^2-1}{x-1}$, and pose $g(1) = a$. Like $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous in 1 if and only if $a = 2$. The point value must therefore be compared with the limit, not replaced by it without justification.

Continuations and continuity. If $u_n \to \ell$ and f is continuous in ℓ, then $f(u_n) \to f(\ell)$. For a recurring sequence convergent $u_{n+1} = f(u_n)$, we get $\ell = f(\ell)$.

Dichotomy

For f continues and $f(a)f(b) < 0$, calculate $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep half of which the extremities have images of opposite signs. n steps, length is $\frac{b-}{2^n}$. The environment The method does not guarantee unity; it proves itself separately.



$y = x^3 - 3x$ and its tangent with inflection.

Linking calculation, curve and argument

4.1 The inside factor and the field

First use: S29.

Example Solved · A two-step composition

For $f(x) = \sqrt{1+x^2}$, the argument is always strictly positive: f is derivable on $\mathbb{R}$. The external function has for derivative $t \mapsto 1/(2\ \bar{t})$, the inner function is derivative 2 x. Their product gives $f'(x) = x/\ \overline{1+x^2}$. The fraction is negative for $x < 0$, positive for $x > 0 : f$ Decline and then grow.

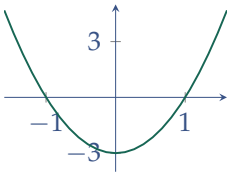
A1 · Immediate application Determine the differentiability domain and derivative of $g(x) = \sqrt{2x-1}$ and then $h(x) = e^{1-x^2}$. Check the sign of g'.

A2 · Repair a derivative "For $f(x) = (3x+1)^4$, we have $f'(x) = 4(3x+1)^3$. » Explain the factor and propose a verification by development or rate of increase in 0.

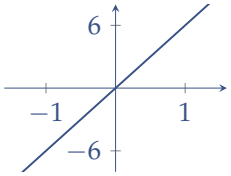
4.2 Associate f, f' and f''

First use: S29 and S31.

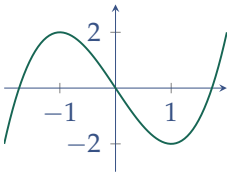
Curve A



Curve B



Curve C



Example resolved · A positive and decreasing function

For $g(x) = e^{-x}$, g is positive, $g' = -e^{-x} < 0$ and $g'' = e^{-x} > 0$. The curve goes down, but its negative slopes increase to 0: it is convex. The three informations are compatible.

A3 · Read before calculating The three curves correspond, in an order to find, to $f(x) = x^3 - 3x$, f' and f''. Associate them using zeros, extrema and changes in convexity; then control by bypass.

Property · Two readings not to be confused

$f'(a) > 0$ describes a positive slope of f, not a positive order. $f''(a) > 0$ indicates locally that the slopes increase; this does not mean that f increase. e^{-x} Decline and stay convex.

4.3 Between a tangent and a rope

First use: S31; take-back S33.

Example resolved · Two frameworks of x^2

On $[0;2]$, the rope joining $(0;0)$ to $(2;4)$ a for equation $y = 2x$; the tangent in 1 has for equation $y = 2x - 1$. Convexity gives 2 $x - 1 \leq x^2 \leq 2x$. Differences are established: $x^2 - (2x - 1) = (x - 1)^2 \geq 0$ and $2x - x^2 = x(2 - x) \geq 0$ over this interval.

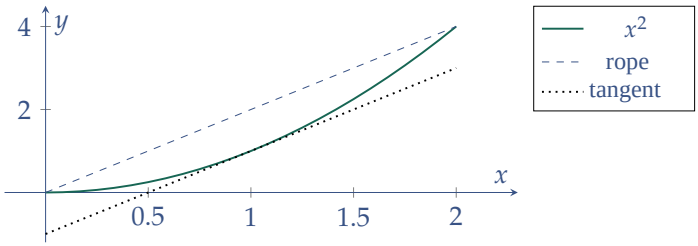


Figure 4.1. Compare the three ordered for $x = \frac{1}{2}$, then justify on all $[0;2]$.

A4 · Compare three functions For x^4, x^3 and $|x|$, study in 0 continuity, differentiability and change for both polynomials, explain why the cancellation of the second derivative does not
It is not enough to conclude.

4.4 Existence and uniqueness are two conclusions

First use: S32 and S33.

Example Solved · Write an Approached Resolution

Let $F(x) = x^3 + x - 1$. It is a polynomial, so it is continuous. $F(0) < 0 < F(1)$ gives a solution in $]0;1[$ by the VIT. $F'(x) = 3x^2 + 1 > 0$ on $\mathbb{R}$ ensures that there can't be two. Calculation $F(0{,}68) < 0 < F(0{,}69)$ locates this solution, but does not replace the existence or Oneness.

A5 · Two counter-examples The function is -1 for $x < 0$ and 1 for $x \geq$ Does it change the sign without cancelling? What hypothesis of the VIT lacks? For $x^2 = 1$, what hypothesis is missing for assert a single solution on $\mathbb{R}$?

4.5 Keep information at each stage

First use: S33 and S40.

Property · Invariant dichotomy

For a continuous function, maintain $F(a)F(b) \leq 0$ keeps at least one zero in $[a;b]$. À each step, the length is divided by 2. The algorithm gives a frame; uniqueness shall be established separately.

Example resolved · A first interval

For $F(x) = x^2 - 2$ on [1;2], $F(1) = -1$, $F(2) = 2$. In the middle $m = 3/2$, $F(m) = 1/4 > 0$; the half that keeps the change of sign is $[1;3/2]$. Its width is 1/2.

A6 · Choice of strategy Demonstrate that $e^{-x} = x$ has a single solution in $[0;1]$. Perform two dichotomy steps, then determine how many steps guarantee a range width at plus 1/128.

A first step: Bring the equation to the cancellation of a function. The number of steps comes from the width, not values of the function.

Wait a minute at the courthouse. Present the existence of a solution in one sentence containing the interval, the continuity and values at the ends; present unity in another sentence.

Review postponed · S47 and then S66 Apply the same arguments to a logarithmic function, then to an area dependent on a bound.

Research activity

Derivatives and graphical readings · 35 min.

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$ and $f''(x)$, then draw up their signboards.
2. Deduct from the changes in f and its intervals of convexity and concavity.
3. Determine the coordinates of the deflection point of $\mathcal{C}_f$ and an equation of tangent at this point.
4. Using the limits, the extrema and the theorem of intermediate values, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure the uniqueness of a solution.